

```
import java.util.Stack;

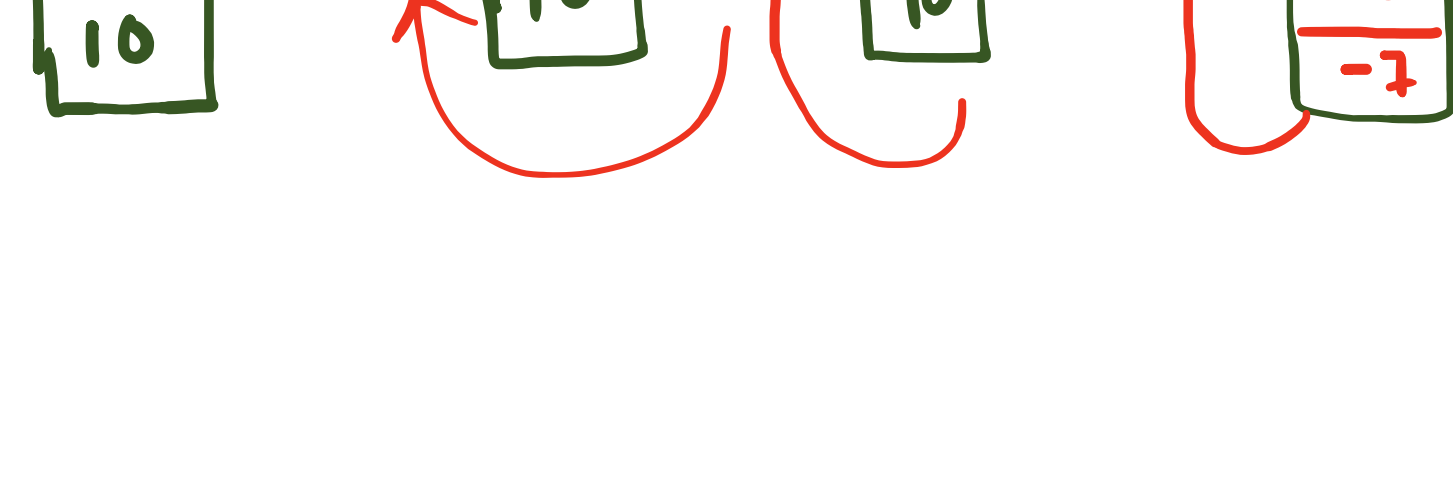
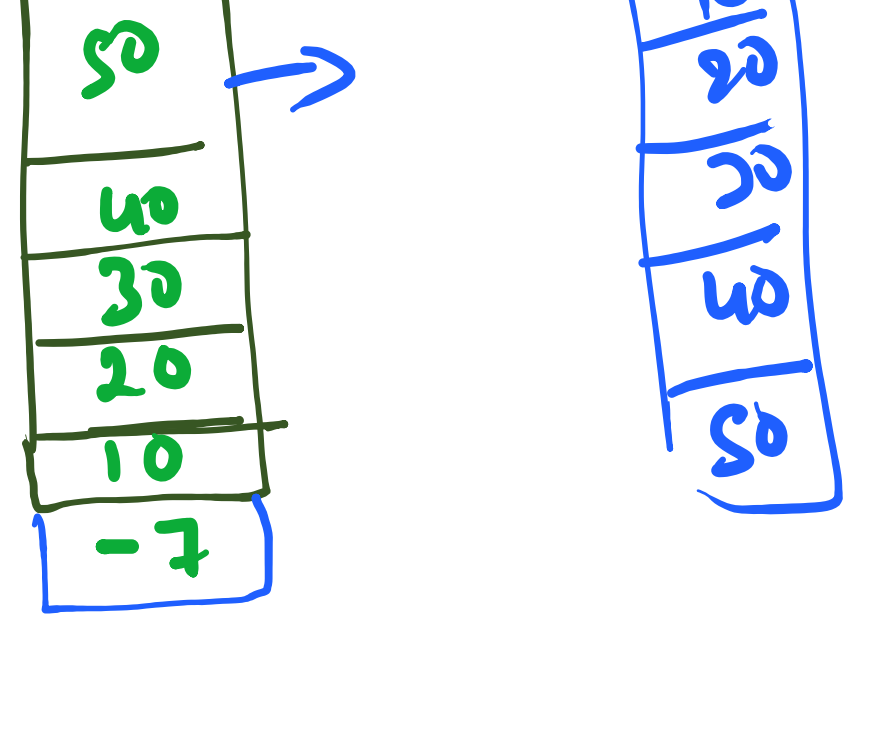
public class Insert_Bottom {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Stack<Integer> st = new Stack<>();
        st.push(10);
        st.push(20);
        st.push(30);
        st.push(40);
        st.push(50);
        System.out.println(st);
        Insert(st, -7);
        System.out.println(st);
    }

    public static void Insert(Stack<Integer> st, int item) {

    }

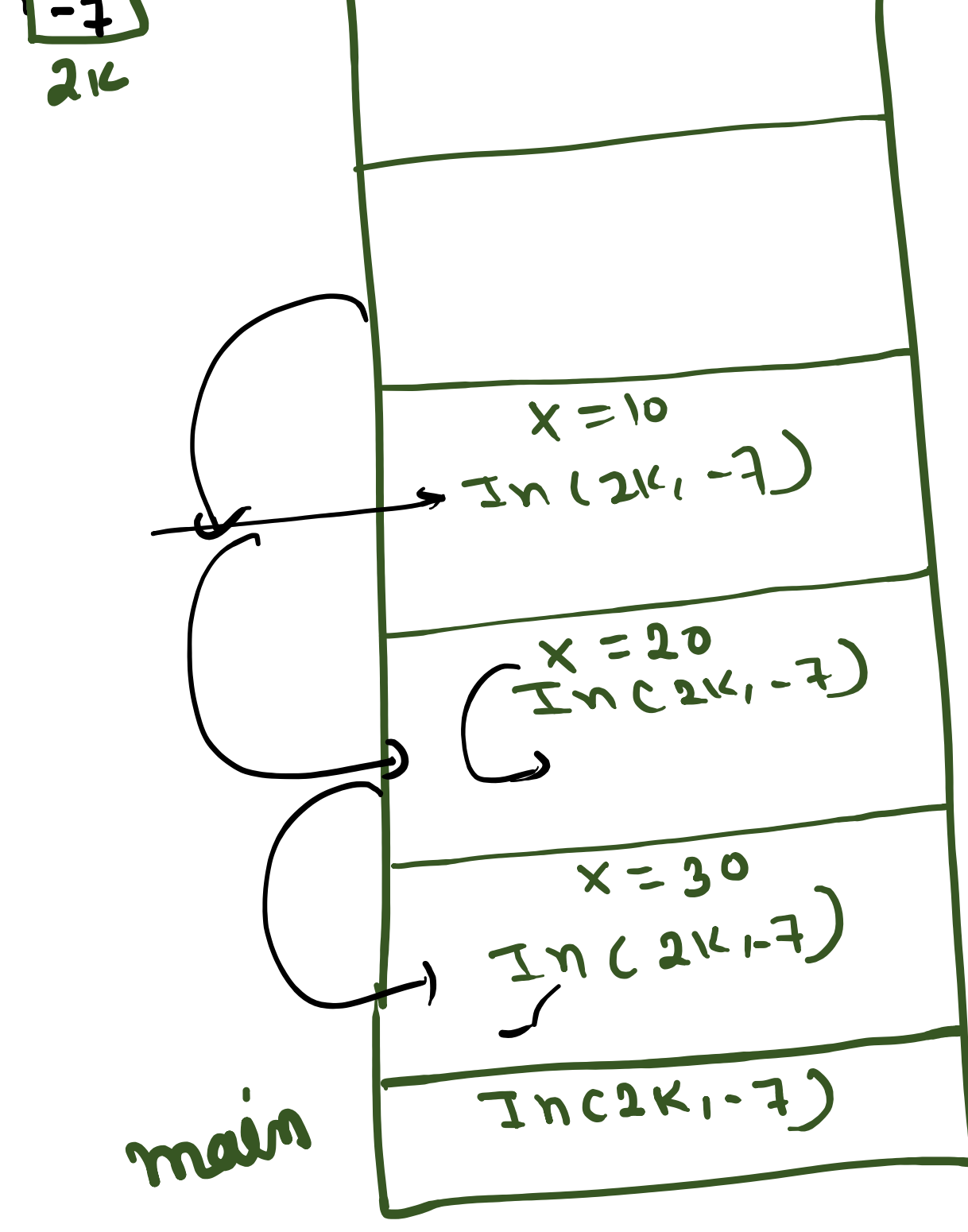
}
```



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    Stack<Integer> st = new Stack<>();
    st.push(10);
    st.push(20);
    st.push(30);
    st.push(40);
    st.push(50);
    System.out.println(st);
    Insert(st, -7);
    System.out.println(st);
}

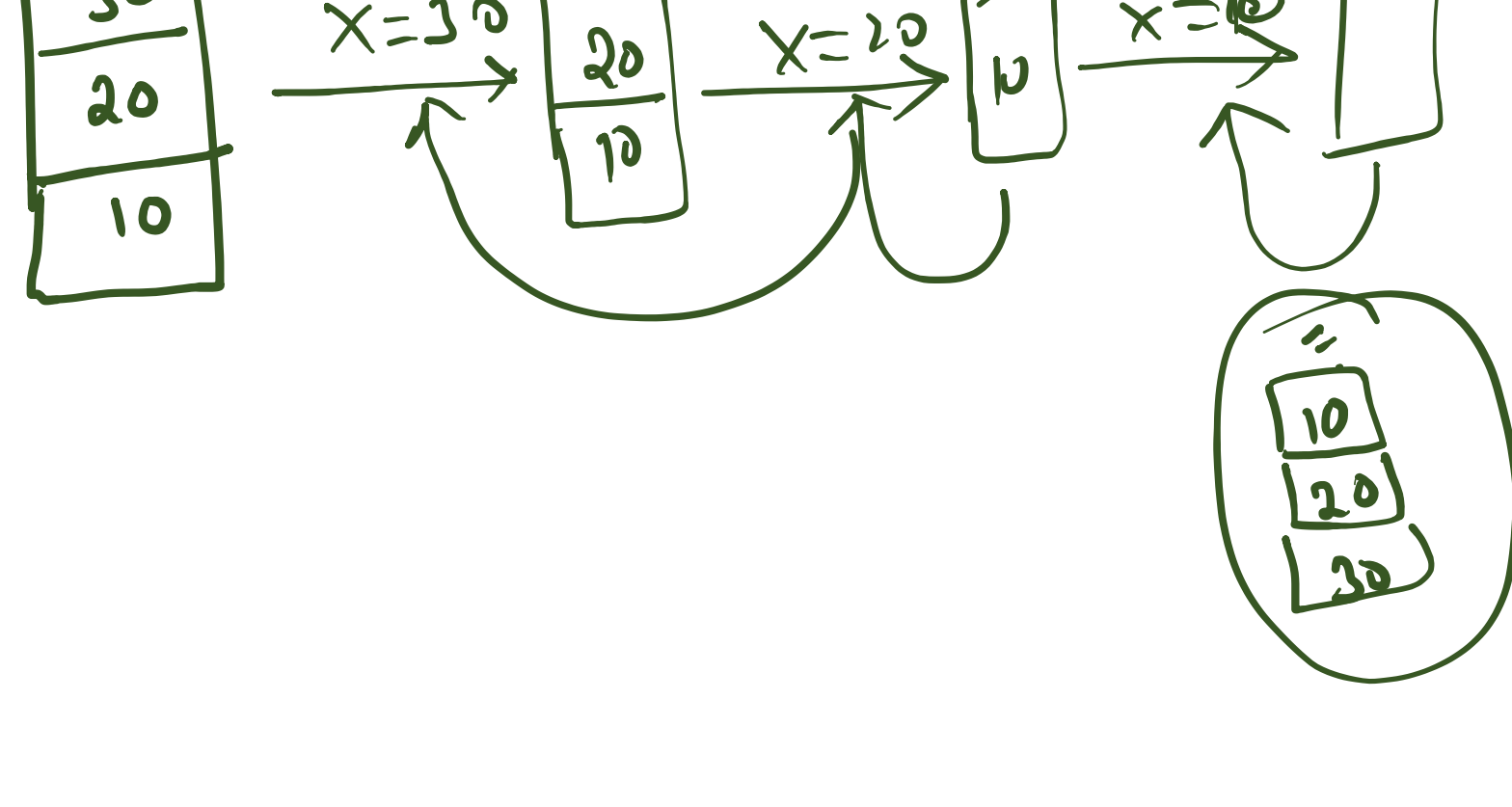
public static void Insert(Stack<Integer> st, int item)
{
    if (st.isEmpty()) {
        st.push(item);
        return;
    }
    int x = st.pop();
    Insert(st, item);
    st.push(x);
}

}
```



```
private static void Reverse(Stack<Integer> st) {
    // TODO Auto-generated method stub
    if (st.isEmpty()) {
        return;
    }
    int x = st.pop();
    Reverse(st);
    st.push(x);
}

}
```



2375. Construct Smallest Number From DI String

Solved

Medium Topics Companies Hint

You are given a 0-indexed string `pattern` of length `n` consisting of the characters `'I'` meaning increasing and `'D'` meaning decreasing.

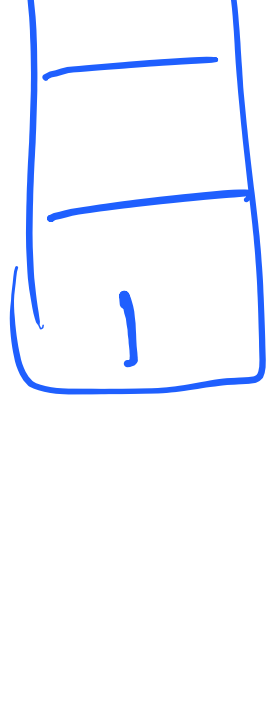
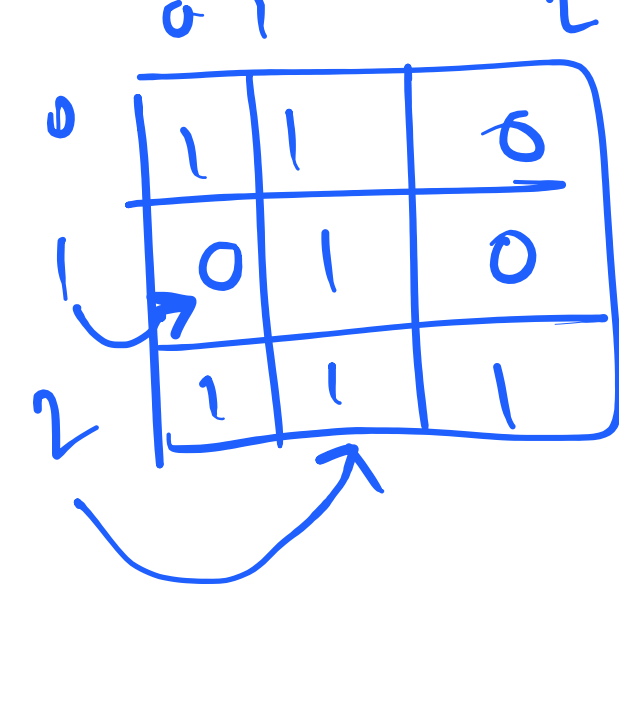
A 0-indexed string `num` of length `n + 1` is created using the following conditions:

- `num` consists of the digits `'1'` to `'9'`, where each digit is used **at most once**.
- If `pattern[i] == 'I'`, then `num[i] < num[i + 1]`.
- If `pattern[i] == 'D'`, then `num[i] > num[i + 1]`.

Return the **lexicographically smallest** possible string `num` that meets the conditions.

Handwritten notes and diagrams for the "Construct Smallest Number From DI String" problem. The notes include a list of conditions, a diagram showing the construction of the number "14328765" from the pattern "I I I I D D D I", and a diagram showing the construction of the number "32165478" from the pattern "D D I D D I". The diagrams show the digits 1 through 9 being placed in the correct order to satisfy the conditions. The notes also include a diagram showing the construction of the number "14328765" from the pattern "I I I I D D D I".

(1,1,0), (0,1,0), (2,1,1)

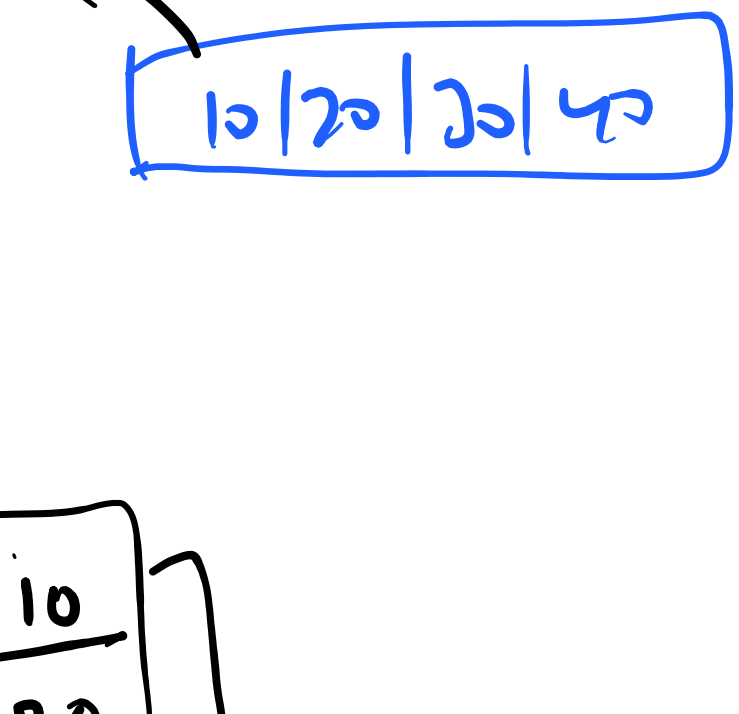
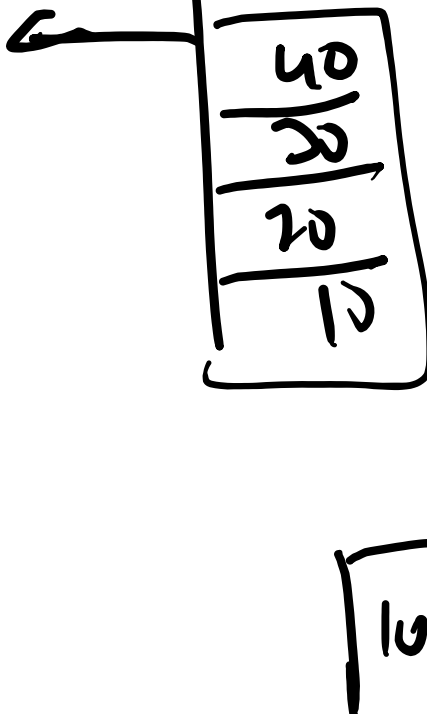
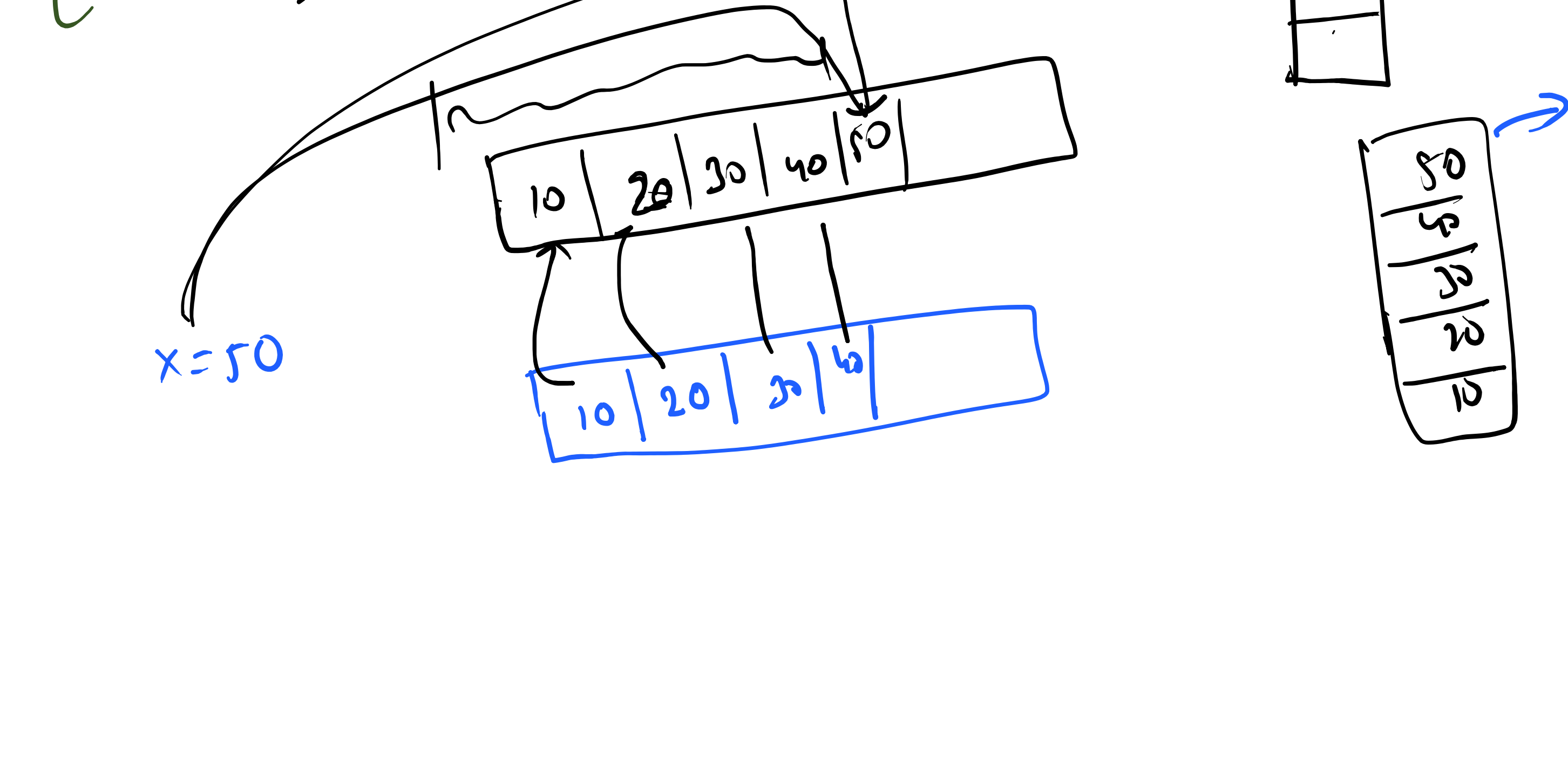


a=2
b=1
a=1
b=0

Given a string `s` containing just the characters `'('`, `'{'`, `'['`, `')`, `'}'`, `']'` and `'`, determine if the input string is valid.

An input string is valid if:

- Open brackets must be closed by the same type of brackets.
- Open brackets must be closed in the correct order.
- Every close bracket has a corresponding open bracket of the same type.



Eq(10)
Eq(20)
Eq(30)
Eq(40)